

5. (c) Let the sum be ₹ x .

$$\text{Then, } \frac{x \times 13 \times 1}{100} - \frac{x \times 12 \times 1}{100} = 110$$

$$\Rightarrow \frac{x}{100} = 110$$

$$\Rightarrow x = 110 \times 100 = ₹ 11000$$

Shortcut Method

Here, $P = ₹ 110$, $r_1 = 13\%$, $r_2 = 12\%$
and $t_1 = t_2 = 1\text{yr}$

∴ Short trick

$$= \frac{110 \times 100}{13 - 12}$$

$$= ₹ 11000$$

6. (b) Let the sum (principal) be ₹ x .

$$\therefore \text{Simple interest} = ₹ \frac{x}{2}$$

and $T = 6\text{ yr}$, $R = 10\%$ per annum

$$\therefore \text{SI} = \frac{P \times R \times T}{100}$$

$$\Rightarrow \frac{x}{2} = \frac{x \times 10 \times 6}{100} \Rightarrow \frac{1}{2} = \frac{6}{10}$$

Which is not true, so it is not a possible case.

7. (c) Let the sum be ₹ 100.

Then, amount = (Sum + SI)

$$= \left(100 + \frac{100 \times 8 \times 5}{100} \right) = (100 + 8 \times 5)$$

So, when the amount is $(100 + 8 \times 5)$, then sum = 100

When the amount is ₹ 840, then sum

$$= ₹ \left(\frac{100 \times 840}{100 + 8 \times 5} \right)$$

8. (b) Simple interest for 4 yr
= ₹ $(1360 - 1040) = ₹ 320$

Simple interest for 3 yr

$$= ₹ \left(\frac{320}{4} \times 3 \right) = ₹ 240$$

∴ Sum or principle

$$= ₹ (1040 - 240) = ₹ 800$$

Shortcut Method

Here, $A_1 = ₹ 1040$, $t_1 = 3\text{ yr}$,

$A_2 = ₹ 1360$ and $t_2 = 7\text{ yr}$

∴ Required principle (sum)

$$= ₹ \left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} \right) \quad [\text{Rule 2}]$$

$$= \frac{1360 \times 3 - 1040 \times 7}{3 - 7} = ₹ 800$$

9. (d) SI at 5% = $6P - P = 5P$

$$\therefore 5P = \frac{P \times 5 \times T}{100} \Rightarrow T = 100\text{ yr}$$

Now, for new rate (R),

$$11P = \frac{P \times R \times 100}{100}$$

$$\therefore R = 11\%$$

Shortcut Method

As, $R_1 = 5\%$, $n_1 = 6$, $n_2 = 12$.

According to the formula,

$$R_2 = \frac{n_2 - 1}{n_1 - 1} \times R_1 = \frac{12 - 1}{6 - 1} \times 5 \quad [\text{Rule 3}]$$

$$= \frac{11}{5} \times 5 = 11\%$$

10. (b) Total borrowed money = ₹ 40000
and rate of interest = 8%

$$\text{The interest for 2 yr} = \frac{40000 \times 8 \times 2}{100}$$

$$= ₹ 6400$$

Let he paid ₹ x at the end of second year.

∴ Interest will be calculated on

$$₹ (40000 - x + 6400)$$

$$\text{Interest for 3 yr} = \frac{(46400 - x) \times 3 \times 8}{100}$$

$$= ₹ \frac{6}{25} (46400 - x)$$

$$\therefore \frac{6}{25} (46400 - x) + 46400 - x = 35960$$

$$\Rightarrow x = \frac{21576 \times 25}{31} = ₹ 17400$$

11. (b) Let sum lent at rate 5% be ₹ x .

Then, sum lent at rate 8%

$$= ₹ (1550 - x)$$

$$\therefore \text{Simple interest at rate 5\%} = \frac{x \times 5 \times 3}{100}$$

$$\text{Simple interest at rate 8\%}$$

$$= \frac{(1550 - x) \times 8 \times 3}{100}$$

$$\therefore \frac{x \times 15}{100} + \frac{(1550 - x) \times 24}{100} = 300$$

$$\Rightarrow \frac{15x}{100} + \frac{37200}{100} - \frac{24x}{100} = 300$$

$$\Rightarrow x = ₹ 800$$

Amount lent at rate 8%

$$= ₹ (1550 - 800) = ₹ 750$$

$$\therefore \text{Required ratio} = \frac{800}{750} = \frac{16}{15} = 16 : 15$$

12. (d) Let the amount remaining to pay be ₹ x .

∴ Price of house = ₹ $(x + 8000)$

$$\Rightarrow 9600 - \frac{x \times 4 \times 5}{100} = x$$

$$\Rightarrow 9600 - \frac{x}{5} = x$$

$$\Rightarrow 9600 = \frac{6x}{5} \Rightarrow \frac{9600 \times 5}{6} = x$$

$$\Rightarrow x = ₹ 8000$$

∴ Cash price of the house

$$= ₹ (8000 + 8000) = ₹ 16000$$

13. (a) Let the principal be ₹ P

As, amount = $3P$ and $T = 25\text{ yr}$

$$\therefore \text{SI} = 3P - P = 2P$$

$$\therefore \text{Rate} = \frac{100 \times \text{SI}}{\text{principal} \times T}$$

$$= \frac{100 \times 2P}{P \times 25} = 8\%$$

14. (a) Let a man invest ₹ 1000 at a $R\%$.

Now, rate is increased by 2%.

$$\therefore \text{New rate} = (R + 2)\%$$

By given condition,

$$\frac{1000 \times R \times 3}{100} + \frac{1500 \times (R + 2) \times 3}{100} = 390$$

$$\Rightarrow 30R + 45R + 90 = 390$$

$$\Rightarrow 75R = 300 \Rightarrow R = 4\%$$

15. (c) Here, rate of interest = $3\frac{1}{8}\% = \frac{25}{8}\%$

Let principal be ₹ x .

$$\text{and simple interest} = ₹ \frac{3}{8} x$$

$$\therefore \frac{3}{8} x = \frac{x \times \frac{25}{8} \times T}{100}$$

$$\Rightarrow \frac{300}{25} = T \Rightarrow T = 12\text{ yr}$$

16. (c) Given, rate of interest = 12% per annum

Simple interest = ₹ 6000 per annum

Let principal is ₹ P .

$$\therefore 6000 = \frac{P \times 1 \times 12}{100}$$

$$\therefore P = \frac{6000 \times 100}{12} = ₹ 50000$$

Hence, the required Principal amount is ₹ 50000.

17. (c) For A Let the amount be ₹ x .

Rate of interest = 5% and Time = 10 yr

$$\therefore \text{Simple interest} = \frac{x \times 5 \times 10}{100} = \frac{x}{2}$$

For B The amount be ₹ $2x$.

Rate of interest = 5% and Time = 5 yr

$$\text{SI} = \frac{2x \times 5 \times 5}{100} = \frac{x}{2}$$

So, A and B both will get the same amount as interest.

18. (a) Let the first part and the second part be ₹ x and ₹ $(625 - x)$, respectively.

∴ The simple interest on both part in same.

$$\text{So, } \frac{x \times 5 \times 2}{100} = \frac{(625 - x) \times 10 \times 4}{100}$$

$$\Rightarrow 10x = (625 - x)40$$

$$\Rightarrow x = (625 - x)4$$

$$\therefore x = ₹ 500$$

$$\therefore \text{Second part} = ₹ (625 - 500) = ₹ 125$$

19. (b) Let the amount of $A = ₹ a$,
time = 2 yr and rate = 5%

∴ Simple Interest of

$$A = \frac{a \times 2 \times 5}{100} = \frac{10a}{100}$$

Let the amount of $B = ₹ b$, rate = 5%
and time = 3 yr.

∴ Simple interest of

$$B = \frac{b \times 3 \times 5}{100} = \frac{15b}{100}$$

Let the amount of $C = ₹ c$, time = 4 yr
and rate = 5%

∴ Simple interest of $C = \frac{c \times 4 \times 5}{100} = \frac{20c}{100}$

$$\text{But } \frac{a \times 10}{100} = \frac{b \times 15}{100} = \frac{c \times 20}{100}$$

$$\Rightarrow 10a = 15b = 20c = k$$

$$\text{So, } a = \frac{k}{10}, b = \frac{k}{15}, c = \frac{k}{20}$$

$$\therefore a : b : c = \frac{1}{10} : \frac{1}{15} : \frac{1}{20}$$

20. (a) Let the sum be $₹ x$ and the original
rate $r\%$, then

$$\text{Simple interest} = \frac{x \times r \times 2}{100}$$

Now, rate is increased by 3%.

∴ New rate = $(r + 3)\%$

$$\therefore \text{Simple interest} = \frac{x \times (r + 3) \times 2}{100}$$

$$\therefore \frac{x \times (r + 3) \times 2}{100} - \frac{x \times r \times 2}{100} = 72$$

$$\Rightarrow \frac{(xr + 3x)2}{100} - \frac{2xr}{100} = 72$$

$$\Rightarrow \frac{2xr + 6x - 2xr}{100} = 72$$

$$\therefore x = ₹ 1200$$

21. (c) As rate of interest is charged half
yearly,

$$\text{So, rate} = \frac{13}{2}\% \text{ half yearly}$$

$$\text{time} \left(\frac{42}{12} \times 2 \right) \text{ half yearly} = 7 \text{ half yearly}$$

$$\text{SI} = \frac{20000 \times 13 \times 7}{100 \times 2} = ₹ 9100$$

$$\therefore \text{Amount (A)} = 20000 + 9100 = ₹ 29100$$

22. (b) Let initial amount be $₹ P$, then
 $A = ₹ 3P$ and

$$T = 15 \text{ yr and } 6 \text{ months} = \frac{31}{2} \text{ yr}$$

$$\therefore \text{SI} = A - P = 3P - P = ₹ 2P$$

$$\Rightarrow P \times \frac{31}{2} \times \frac{R}{100} = 2P$$

$$\Rightarrow R = \frac{2 \times 2 \times 100}{31}\% = \frac{400}{31}\%$$

Let amount doubled be t_1 yr.

Now, $\text{SI} = 2P - P = ₹ P$

$$\therefore t_1 = \frac{\text{SI} \times 100}{P \times R}$$

$$\Rightarrow t_1 = \frac{P \times 100 \times 31}{P \times 400} = \frac{31}{4} = 7 \text{ yr and } 9 \text{ months}$$

23. (c) Case I $\text{SI} = \frac{P \times R \times T}{100}$

$$R = \frac{24200 \times 4 \times 6}{100} = ₹ 5808$$

∴ Amount = Principal + SI

$$\text{SI} = 24200 + 5808 = 30008$$

$$\text{Case II } \text{SI} = \frac{30008 \times 4 \times 4}{100} = ₹ 4801.28$$

24. (d) Let the person invest amount x and y
into two different rates of interest.

$$\therefore \frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100} = 130$$

$$\left[\because \text{SI} = \frac{\text{PRT}}{100} \right]$$

$$\Rightarrow 12x + 10y = 13000 \quad \dots(i)$$

$$\text{and } \frac{y \times 12 \times 1}{100} + \frac{x \times 10 \times 1}{100} = 134$$

$$\Rightarrow 12y + 10x = 13400 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$x = ₹ 500 \text{ and } y = ₹ 700$$

25. (d) Here, $I_1 = 50$, $I_2 = 80$, $T_1 = 4$ yr,
 $T_2 = 8$ yr, and $P = ₹ 1500$

According to formula,

$$I_2 - I_1 = \frac{P \times R \times (T_2 - T_1)}{100}$$

$$\Rightarrow 80 - 50 = \frac{1500 \times R \times (8 - 4)}{100}$$

$$\Rightarrow 30 = 15 \times R \times 4$$

$$\Rightarrow 2R = 1$$

$$\Rightarrow R = \frac{1}{2} = 0.5\%$$

26. (a) Here, $I_1 - I_2 = ₹ 10$, $P_1 = ₹ 1230$,

$$P_2 = ₹ 1130 \text{ and } T = 2 \text{ yr}$$

According to formula,

$$I_2 - I_1 = \frac{T \times R \times (P_2 - P_1)}{100}$$

$$-10 = \frac{R \times 2(1130 - 1230)}{100}$$

$$-10 = \frac{-200R}{100}$$

$$\therefore R = 5\%$$

27. (b) Case I Annual payment

$$= \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{1092 \times 100}{100 \times 2 + \frac{24(2-1)}{2}}$$

$$= \frac{1092 \times 100}{212} = ₹ 515.09 \approx ₹ 515$$

Case II Annual payment

$$= \frac{1092 \times 100}{100 \times 2 + \frac{12(2-1)}{2}}$$

$$= \frac{1092 \times 100}{206} = ₹ 530.09 \approx ₹ 530$$

28. (c) Let P be the principal amount.

$$\text{SI} = ₹ 55, \text{ time, } t = 9 \text{ months} = \frac{9}{12} \text{ yr}$$

$$\text{and rate, } r = 3\frac{2}{3}\% = \frac{11}{3}\%$$

$$\therefore \text{SI} = \frac{P \times R \times T}{100}$$

$$\Rightarrow P = \frac{\text{SI} \times 100}{R \times T} = \frac{55 \times 100}{\frac{11}{3} \times 9} \times 3 \times 12$$

$$= 5 \times 100 \times 4 = ₹ 2000$$

$$\therefore \text{Principal (P)} = ₹ 2000$$

29. (a) Let the principal be $₹ x$ and the time
be t yr.

Rate = 10%

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100}$$

According to the question,

$$0.125 \times \text{Principal} = \text{Simple interest}$$

$$\therefore 0.125 P = \frac{P \times 10 \times T}{100}$$

$$\Rightarrow \frac{125P}{1000} = \frac{10 \times P \times T}{100}$$

$$\Rightarrow \frac{125}{100} = T$$

$$\Rightarrow T = \frac{5}{4} = 1\frac{1}{4} \text{ yr}$$

30. (c) Let principal = $₹ P$, then amount of
money = $₹ 2P$

$$\therefore \text{SI} = 2P - P = ₹ P$$

$$\text{Now, } P = \frac{P \times r \times 5}{100} \Rightarrow r = 20\%$$

Amount of money after 12 yr = $₹ 3P$

$$\therefore \text{SI} = 3P - P = 2P$$

$$\text{Now, } 2P = \frac{P \times R \times 12}{100} \Rightarrow R = \frac{50}{3}\%$$

∴ Difference between two interest rates

$$= \left(20 - \frac{50}{3} \right)\% = \frac{10}{3}\% = 3\frac{1}{3}\%$$

10

COMPOUND INTEREST

Usually (1-2) questions have been asked from this chapter. Questions from this chapter are based on the direct application of compound interest formula.



COMPOUND INTEREST

When the interest is calculated on the amount of previous year, then it is known as compound interest. It will make a deposit or loan grow at a faster rate than simple interest.

Important Rules and Formulae

Consider P = Principal (amount borrowed), A = Amount (Principal + Interest)
 R = Rate of interest per annum and n = Number of years

Rule 1 When interest is compounded annually, then

$$(i) \text{ Amount} = P \left(1 + \frac{R}{100}\right)^n \quad (ii) \text{ CI} = A - P = P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right]$$

EXAMPLE 1. The amount and the compound interest on ₹ 24000 compounded annually for 2 yr at the rate of 10% per annum is

- a. ₹ 39040 and ₹ 4040 b. ₹ 29040 and ₹ 5040 c. ₹ 19040 and ₹ 3040 d. None of these

Sol. b. Here, $P = ₹ 24000$, $R = 10\%$ per annum and $n = 2$ yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n = 24000 \left(1 + \frac{10}{100}\right)^2 = 24000 \left(\frac{110}{100}\right)^2 = 24000 \times \frac{110}{100} \times \frac{110}{100} = ₹ 29040$$

$$\therefore \text{Compound interest} = \text{Amount} - \text{Principal} = ₹ 29040 - ₹ 24000 = ₹ 5040$$

So, amount = ₹ 29040 and compound interest = ₹ 5040

Rule 2 When interest is compounded half-yearly (every 6 months), then

$$(i) \text{ Amount} = P \left(1 + \frac{r}{100 \times 2}\right)^{2n} \quad (ii) \text{ CI} = \left[\left(1 + \frac{R}{100 \times 2}\right)^{2n} - 1 \right]$$

EXAMPLE 2. The compound interest on ₹ 24000 compounded semi-annually for $1\frac{1}{2}$ yr at the rate of 10% per annum is

- a. ₹ 3783 b. ₹ 2783 c. ₹ 2763 d. ₹ 3763

Sol. a. Given, $P = ₹ 24000$

Rate of interest = 10% per annum

and time = $1\frac{1}{2}$ yr = $\frac{3}{2}$ yr

Since, the interest is compounded half-yearly.

Then, rate $(R) = \frac{10}{2}\%$ half-yearly

and time $(n) = 2 \times \frac{3}{2} = 3$ half-yearly

$$\begin{aligned} \therefore A &= 24000 \left(1 + \frac{5}{100}\right)^3 \quad \left[\because A = P \left(1 + \frac{R}{100}\right)^n\right] \\ &= 24000 \left(1 + \frac{1}{20}\right)^3 = 24000 \left(\frac{21}{20}\right)^3 \quad [\text{Rule 1}] \\ &= 24000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} = ₹ 27783 \end{aligned}$$

Now, compound interest = $A - P$

$$= ₹ 27783 - ₹ 24000 = ₹ 3783$$

(Rule 3) When interest is compounded quarterly (every 3 months), then

$$(i) \text{ Amount} = P \left(1 + \frac{R}{100 \times 4}\right)^{4n}$$

$$(ii) \text{ CI} = P \left[\left(1 + \frac{R}{100 \times 4}\right)^{4n} - 1 \right]$$

EXAMPLE 3. The amount and the compound interest on ₹ 100000 compounded quarterly for 9 months at the rate of 4% per annum is

- a. ₹ 106060 and ₹ 6060.10 b. ₹ 103030 and ₹ 3030.10
c. ₹ 103030.10 and ₹ 3030.10 d. ₹ 106060.10 and ₹ 6060.10

Sol. c. Given, $P = ₹ 100000$, rate of interest = 4% per annum and time = 9 months

Since, the interest is compounded quarterly.

Then, rate $(R) = \frac{4}{4} = 1\%$ quarterly

and time $(n) = 9 \text{ months} = \frac{9}{12} \times 4 = 3$ quarter yearly

$$\therefore \text{Amount, } A = P \left(1 + \frac{R}{100}\right)^n$$

$$\therefore A = 100000 \left(1 + \frac{1}{100}\right)^3 = 100000 \left(\frac{101}{100}\right)^3$$

$$= 100000 \times \frac{101}{100} \times \frac{101}{100} \times \frac{101}{100} = ₹ 103030.10$$

$$\therefore \text{Compound interest} = A - P = ₹ 103030.10 - ₹ 100000 = ₹ 3030.10$$

(Rule 4) When interest is compounded annually but time is given in fraction (say, $t\frac{a}{b}$ yr), then

$$\text{Amount} = P \left(1 + \frac{R}{100}\right)^t \times \left(\frac{1 + (a/b)R}{100}\right)$$

EXAMPLE 4. The amount and compound interest on ₹ 5000 compounded annually for 2 yr 6 months at the rate of 10% per annum is

- a. ₹ 6252.5 and ₹ 1342.5 b. ₹ 6000 and ₹ 1300
c. ₹ 6250 and ₹ 1340 d. ₹ 6352.5 and ₹ 1352.5

Sol. d. Here, time = 2 yr and 6 months

$$\therefore t = 2 \text{ yr and } \frac{a}{b} = \frac{1}{2} \text{ yr}$$

$$\therefore \text{Amount} = P \left(1 + \frac{R}{100}\right)^t \times \left(\frac{1 + \frac{a}{b}R}{100}\right)$$

$$\therefore A = 5000 \left(1 + \frac{10}{100}\right)^2 \left(1 + \frac{\frac{1}{2} \cdot 10}{100}\right)$$

$$= 5000 \left(\frac{110}{100}\right)^2 \left(\frac{105}{100}\right)$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{21}{20} = ₹ 6352.50$$

$$\therefore \text{Compound interest} = A - P$$

$$\text{CI} = 6352.50 - 5000 = ₹ 1352.50$$

(Rule 5) When rate of interest for n_1 , n_2 and n_3 yr are R_1 , R_2 and R_3 respectively, then

$$\text{Amount} = P \left(1 + \frac{R_1}{100}\right)^{n_1} \left(1 + \frac{R_2}{100}\right)^{n_2} \left(1 + \frac{R_3}{100}\right)^{n_3}$$

EXAMPLE 5. The compound interest on ₹ 5000 for 4 yr if the rate of interest is 10% per annum for the first two years and 15% for the next two years is

- a. ₹ 3000 b. ₹ 3001 c. ₹ 3002 d. None of these

Sol. b. Here, $R_1 = 10\%$, $n = n_1 + n_2 = 4$ yr

$$n_1 = 2 \text{ yr, } R_2 = 15\%, n_2 = 2 \text{ yr}$$

and principal = ₹ 5000

$$\therefore \text{Amount} = P \left(1 + \frac{R_1}{100}\right)^{n_1} \left(1 + \frac{R_2}{100}\right)^{n_2}$$

$$\therefore A = 5000 \left(1 + \frac{10}{100}\right)^2 \left(1 + \frac{15}{100}\right)^2$$

$$= 5000 \times \left(\frac{110}{100}\right)^2 \left(\frac{115}{100}\right)^2$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{23}{20} \times \frac{23}{20} = ₹ 8001.125$$

$$\therefore \text{Compound interest}$$

$$= A - P = ₹ 8001.125 - ₹ 5000 \approx ₹ 3001$$

Rule 6 The difference D between simple and compound interest accrued on ₹ P at the rate of interest of $R\%$ is given by

$$(i) \text{ For 2 yr, } D = \frac{PR^2}{(100)^2}$$

$$(ii) \text{ For 3 yr, } D = \frac{PR^2(300+r)}{(100)^3}$$

EXAMPLE 6. The difference between the compound interest and the simple interest on a certain sum at 12% per annum for 2 yr is ₹ 90. Then, sum is

- a. ₹ 6250 b. ₹ 6350 c. ₹ 6520 d. ₹ 6950

Sol. a. Let the sum be ₹ 100.

Case I Then, simple interest on ₹ 100 for 2 yr

$$= \left(\frac{100 \times 12 \times 2}{100} \right) = ₹ 24$$

Case II Amount when ₹ 100 is borrowed for 2 yr

$$= 100 \times \left(1 + \frac{12}{100} \right)^2 \left[A = P \left(1 + \frac{R}{100} \right)^n, \text{Rule 1 (i)} \right]$$

$$= 100 \times \frac{28}{25} \times \frac{28}{25} = ₹ \left(\frac{3136}{25} \right)$$

$$\therefore \text{Compound interest for 2yr} = \left(\frac{3136}{25} - 100 \right) = ₹ \left(\frac{636}{25} \right)$$

$$\text{Difference between CI and SI} = \left(\frac{636}{25} - 24 \right) = ₹ \frac{36}{25}$$

If the difference between CI and SI is ₹ $\frac{36}{25}$, then the sum = ₹ 100

If the difference between CI and SI is ₹ 1,

$$\text{then the sum} = ₹ \left(\frac{100 \times 25}{36} \right)$$

If the difference between CI and SI is ₹ 90, then the sum

$$= \left(100 \times \frac{25}{36} \times 90 \right) = ₹ 6250$$

Here, the sum is ₹ 6250.

Shortcut Method

Let the required sum be ₹ P .

Here, difference = ₹ 90 and $R = 12\%$

$\therefore$ Difference between CI and SI

$$90 = \frac{PR^2}{(100)^2} = \frac{(P)(12)^2}{(100)^2} \Rightarrow 90 = \frac{P \times 144}{100 \times 100}$$

$$\therefore P = ₹ 6250$$

Required sum = ₹ 6250

Rule 7 If a certain sum at compound interest becomes A_1 , in n yr and A_2 in $(n+1)$ yr, then

$$(i) \text{ Rate of compound interest} = \frac{(A_2 - A_1)}{A_1} \times 100\%$$

$$(ii) \text{ Sum} = A_1 \left(\frac{A_2}{A_1} \right)^n$$

EXAMPLE 7. A sum of money on compound interest amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr. What is the rate of interest per annum?

- a. 5% b. 10% c. 15% d. 20%

Sol. b. Given, $A_1 = ₹ 9680$ and $A_2 = ₹ 10648$

$$\therefore \text{Rate of compound interest} = \frac{(A_2 - A_1)}{A_1} \times 100\%$$

$$= \frac{10648 - 9680}{9680} \times 100\% = 10\%$$

DEPRECIATION

The value of certain things like the machine, vehicle etc., decreases over a period of time. The decrement in the value things is called depreciation. The depreciation per unit time is called the rate of depreciation.

Rule 8 If the present value of a article is P and it depreciate at the rate of $R\%$ per annum. Then,

$$(i) \text{ Value of article after } n \text{ yr} = P \times \left(1 - \frac{R}{100} \right)^n$$

$$(ii) \text{ Value of article } n \text{ yr ago} = \frac{P}{\left(1 - \frac{R}{100} \right)^n}$$

EXAMPLE 8. Given that carbon - 14 (C_{14}) decays at a constant rate in such a way that it reduces to 50% in 5568 yr. Then, the age of an old wooden piece in which the carbon is only 12.5% of the original is

- a. 16000 yr b. 16244 yr c. 16702 yr d. None of these

Sol. d. Let the rate of decay be $R\%$ and the age of the wooden piece be n yr.

$$\text{Then, } \left(\frac{50}{100} \right) = \left(1 - \frac{R}{100} \right)^{5568} \Rightarrow \left(1 - \frac{R}{100} \right) = (0.5)^{1/5568} \dots (i)$$

$$\Rightarrow \frac{12.5}{100} = \left(1 - \frac{R}{100} \right)^n \Rightarrow (0.125) = \left(1 - \frac{R}{100} \right)^n$$

$$\Rightarrow (0.125)^{1/n} = \left(1 - \frac{R}{100} \right) \Rightarrow (0.5)^{3/n} = \left(1 - \frac{R}{100} \right) \dots (ii)$$

$$\text{From Eqs. (i) and (ii), } (0.5)^{3/n} = (0.5)^{\frac{1}{5568}}$$

On comparing both sides, we get

$$\frac{3}{n} = \frac{1}{5568} \Rightarrow n = 3 \times 5568 = 16704 \text{ yr}$$

INSTALMENTS

When a borrower pays the sum in parts, then we say that he/she is paying in instalments.

$$\therefore P = \frac{x}{\left(1 + \frac{R}{100} \right)} + \frac{x}{\left(1 + \frac{R}{100} \right)^2} + \dots + \frac{x}{\left(1 + \frac{R}{100} \right)^n}$$